GMAT Quantitative Reasoning

Reference Sheet

Study Material

Quick reference guide for essential formulas, rules, and concepts.

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EDUCATIONAL PURPOSE: This material is intended solely for educational purposes to help students prepare for standardized tests.

1. Arithmetic and Decimals

Absolute Value

$|x|$ is x if $x \geq 0$ and $-x$ if $x < 0$.

For any x and y:
- $|x+y| \leq |x| + |y|$
- $|xy| = |x| \cdot |y|$
- $\sqrt{x^2} = |x|$

Even and Odd Numbers

Operation	Result
Even $\times$ Even	Even
Even $\times$ Odd	Even
Odd $\times$ Odd	Odd
Even $+$ Even	Even
Odd $+$ Odd	Even
Even $+$ Odd	Odd

Addition & Subtraction

$$x + 0 = x = x - 0$$

$$x - x = 0$$

$$x + y = y + x$$

$$x - y = -(y - x) = -y + x$$

$$(x + y) + z = x + (y + z)$$

If x and y are both positive, then $x + y$ is positive.

If x and y are both negative, then $x + y$ is negative.

Multiplication & Division

$$x \times 1 = x = \frac{x}{1}$$

$$x \times 0 = 0$$

$$\text{If } x \neq 0, \text{ then } \frac{x}{x} = 1$$

$$\frac{x}{0} \text{ is undefined}$$

$$xy = yx$$

If $x \neq 0$ and $y \neq 0$, then $\frac{x}{y} = \frac{1}{\frac{y}{x}}$.

$(xy)z = x(yz)$

$xy + xz = x(y + z)$

If $y \neq 0$, then $\left(\frac{x}{y}\right) + \left(\frac{z}{y}\right) = \frac{x+z}{y}$.

Quotients and Remainders

The quotient q and the remainder r of dividing positive integer x by positive integer y are unique positive integers such that:

$$y = xq + r \text{ and } 0 < r < x$$

The remainder r is 0 if and only if y is divisible by x. Then x is a factor of y.

Decimals

Adding/Subtracting: Line up decimal points

$$\begin{array}{rr} & 17.6512 \quad 653.2700 \\ 1. & 653.2700 \quad -17.6512 \\ & 670.9212 \quad 635.6188 \end{array}$$

Multiplying decimal A by decimal B:

Multiply as integers. If A has n digits right of decimal and B has m digits right of decimal, place decimal point in $A \times B$ so it has $m + n$ digits to its right.

Dividing: Move decimal points equally many digits right until B is an integer, then divide.

Scientific Notation

To convert $A \times 10^n$ to regular notation, move A's decimal point n places to the right if n is positive, or $|n|$ places to the left if n is negative.

To convert a decimal to scientific notation, move the decimal point n spaces so exactly one nonzero digit is to its left. Multiply by 10^n if you moved left or by 10^{-n} if moved right.

Divisibility Rules

An integer is divisible by:

Divisor	Rule
2	Final digit is 0, 2, 4, 6, or 8
3	Sum of digits is divisible by 3
4	Final two digits make a number divisible by 4
5	Final digit is 5 or 0
9	Sum of digits is divisible by 9
10	Final digit is 0

2. Exponents

Squares, Cubes, and Square Roots

Every positive number has two real square roots, one positive and one negative.

n	n^2	n^3	$\sqrt{n}$
1	1	1	1
2	4	8	1.41
3	9	27	1.73
4	16	64	2
5	25	125	2.24
6	36	216	2.45
7	49	343	2.65
8	64	512	2.83
9	81	729	3
10	100	1,000	3.16

Exponentiation Rules

$$x^1 = x$$

$$x^0 = 1 \text{ (if } x \neq 0 \text{)}$$

$$x^{-1} = \frac{1}{x} \text{ (if } x \neq 0 \text{)}$$

$$x^y > x \text{ (if } x > 1 \text{ and } y > 1 \text{)}$$

$$x^y < x \text{ (if } 0 < x < 1 \text{ and } y > 1 \text{)}$$

$$(x^y)^z = x^{yz} = (x^z)^y$$

$$x^{y+z} = x^y x^z$$

$$x^{y-z} = \frac{x^y}{x^z} \text{ (if } x \neq 0 \text{)}$$

$$(xy)^z = x^z y^z$$

$$\left(\frac{x}{y}\right)^z = \frac{x^z}{y^z} \text{ (if } y \neq 0 \text{)}$$

Examples:

- $2^1 = 2$
- $2^0 = 1$
- $2^{-1} = \frac{1}{2}$
- $2^3 = 8 > 2$
- $0.2^3 = 0.008 < 0.2$
- $(2^3)^4 = 2^{12} = (2^4)^3$
- $2^7 = 2^3 2^4$
- $2^{5-3} = \frac{2^5}{2^3}$
- $6^4 = 2^4 3^4$
- $\left(\frac{3}{4}\right)^2 = \frac{3^2}{4^2} = \frac{9}{16}$

3. Algebraic Expressions and Linear Equations

Translating Words into Math Operations

$x+y$	$x-y$	xy	$\frac{x}{y}$	x^y
x added to y	x decreased by y	x multiplied by y	x divided by y	x to the power of y
x increased by y	difference of x and y	product of x and y	x over y	x to the yth power
x more than y	y fewer than x	x times y	quotient of x and y	
x plus y	y less than x		ratio of x to y	
sum of x and y	x minus y	If $y=2$: double x, twice x	If $y=2$: half of x, x halved	If $y=2$: x squared
total of x and y	x reduced by y	If $y=3$: triple x		If $y=3$: x cubed
	y subtracted from x			

Manipulating Expressions

Factor to combine like terms:

$3xy-9y=3y(x-3)$

Divide out common factors:

$\frac{3xy-9y}{x-3}=\frac{3y(x-3)}{x-3}=3y(1)=3y$

Multiply expressions (FOIL):

$(3x-4)(9y+x) \quad =3x(9y+x)-4(9y+x) \quad = 3x(9y)+3x(x)-4(9y)-4(x) \quad =27xy+3x^2-36y-4x$

Substitute constants:

If $x=3$ and $y=-2$, then $3xy-x^2+y=3(3)(-2)-(3)^2+(-2)=-18-9-2=-29$

Solving Linear Equations

One variable:

Isolate variable by doing same operations on both sides.

Example: $\frac{5x-6}{3}=4$

1. Multiply both sides by 3: $5x-6=12$
2. Add 6 to both sides: $5x=18$
3. Divide both sides by 5: $x=\frac{18}{5}$

Two variables (x and y):

Solve $x-y=2$ and $3x+2y=11$:

1. From first equation: $x=2+y$
2. Substitute in second: $3(2+y)+2y=11$ $6+3y+2y=11$ $5y=5$, so $y=1$
3. From first: $x=2+1=3$

4. Factoring, Quadratic Equations, and Inequalities

Factoring Formulas

$$\left(a+\frac{1}{a}\right)^2 = a^2 + \frac{1}{a^2} + 2$$

$$a^2 - b^2 = (a-b)(a+b)$$

$$a^2 + 2ab + b^2 = (a+b)^2$$

$$a^2 - 2ab + b^2 = (a-b)^2$$

$$a^2 + b^2 = (a+b)^2 - 2ab = (a-b)^2 + 2ab$$

$$(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$$

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

The Quadratic Formula

For $ax^2 + bx + c = 0$ with $a \neq 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

These roots are two distinct real numbers if $b^2 - 4ac \geq 0$.

If $b^2 - 4ac = 0$, one root: $-\frac{b}{2a}$.

If $b^2 - 4ac < 0$, no real roots.

Solving by Factoring

Steps:
1. Start with polynomial equation
2. Add/subtract to get 0 on one side
3. Write nonzero side as product of factors
4. Set each factor to 0 to find solutions

Example: $x^3 - 2x^2 + x = -5(x-1)^2$

$$x^3 - 2x^2 + x + 5(x-1)^2 = 0$$

$$x(x^2 - 2x + 1) + 5(x-1)^2 = 0$$

$$(x+5)(x-1)^2 = 0$$

So $x + 5 = 0$ or $x - 1 = 0$, giving $x = -5$ or $x = 1$.

Solving Inequalities

As in equations, same operations on both sides. But **multiplying or dividing by a negative reverses the inequality**.

Thus, $6 > 2$, but $(-1)(6) < (-1)(2)$.

Example: $\frac{5x-1}{2} < 3$

1. Multiply by −2 (reversing inequality): $5x - 1 > -6$
2. Add 1: $5x > -5$
3. Divide by 5: $x > -1$

5. Lines in the Coordinate Plane

Line Equations

An equation $y = mx + b$ defines a line with slope m and y-intercept b.

For a line through (x_1, y_1) and (x_2, y_2) with $x_1 \neq x_2$, the slope is:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Given point (x_1, y_1) and slope m, any point (x, y) on the line satisfies:

$$m = \frac{y - y_1}{x - x_1}$$

Finding Line Equation - Example

Given points $(3, -3)$ and $(-2, 4)$, the slope is:

$$m = \frac{(-3) - 4}{3 - (-2)} = -\frac{7}{5}$$

Using point $(3, -3)$:

$$y - (-3) = \left(-\frac{7}{5}\right)(x - 3)$$

$$y + 3 = \left(-\frac{7}{5}\right)x + \frac{21}{5}$$

$$y = \left(-\frac{7}{5}\right)x + \frac{6}{5}$$

So y-intercept is $\frac{6}{5}$.

For x-intercept, set $y = 0$:

$$0 = \left(-\frac{7}{5}\right)x + \frac{6}{5}$$

$$\left(\frac{7}{5}\right)x = \frac{6}{5}$$

$$x = \frac{6}{7}$$

6. Rates, Ratios, and Percents

Fractions

Equivalent Fractions:

Divide numerator and denominator by their GCD.

Operations:

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$$

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

$$\frac{\frac{a}{b}}{\frac{c}{d}} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

Mixed Numbers:

$a\frac{b}{c} = \frac{ac+b}{c}$

Percents

$$x\% = \frac{x}{100}$$

$$x\% \text{ of } y \text{ equals } \frac{xy}{100}$$

To convert percent to decimal, drop %, move decimal point 2 digits left.

To convert decimal to percent, add %, move decimal point 2 digits right.

Percent increase from x to y: $100\left(\frac{y-x}{x}\right)\%$

Percent decrease from x to y: $100\left(\frac{x-y}{x}\right)\%$

Discounts

A price discounted by n percent becomes $(100-n)$ percent of original.

A price discounted by n percent and then by m percent becomes $(100-n)(100-m)$ percent of original.

Rate Problems

$$\text{distance} = \text{rate} \times \text{time}$$

$$\text{Profit} = \text{Revenues} - \text{Expenses}$$

$$\text{Profit} = \text{Selling price} - \text{Cost}$$

Interest

Simple interest:

$$I = P \times r \times t$$

where P is principal, r is interest rate, t is time.

Compound interest over n periods:

$$A = P \times (1+r)^n$$

Work

$$\frac{1}{r} + \frac{1}{s} = \frac{1}{h}$$

where r is how long one individual takes, s is how long a second takes, and h is how long they take together.

Mixtures

	Number of units	Amount per unit	Total amount
Substance A	X	M	X × M
Substance B	Y	N	Y × N
Mixture	X + Y	$\frac{(X \times M)+(Y \times N)}{X+Y}$	(X × M) + (Y × N)

7. Statistics, Sets, and Counting Methods

Statistics

Concept	Definition for n values	Example: {4, 4, 5, 7, 10}
Mean	Sum of n values, divided by n	$\frac{4+4+5+7+10}{5} = \frac{30}{5} = 6$
Median	Middle value if n odd; mean of two middle if n even	5 is the middle value
Mode	Value appearing most often	4 appears twice (only value appearing >1)
Range	Largest value minus smallest	$10 - 4 = 6$
Standard Deviation	1. Find mean 2. Find differences from mean 3. Square each difference 4. Find average of squared differences 5. Take nonnegative square root	(1) Mean is 6 (2) −2, −2, −1, 1, 4 (3) 4, 4, 1, 1, 16 (4) $\frac{26}{5} = 5.2$ (5) $\sqrt{5.2}$

Sets

Concept	Notation	Example
Number of elements	$\lvert S \rvert$	$S = \{-5, 0, 1\}$ has $\lvert S \rvert = 3$
Subset	$S \subseteq T$ (S is subset of T)	$\{-5, 0, 1\}$ is subset of $\{-5, 0, 1, 4, 10\}$
Union	$S \cup T$	$\{3, 4\} \cup \{4, 5, 6\} = \{3, 4, 5, 6\}$
Intersection	$S \cap T$	$\{3, 4\} \cap \{4, 5, 6\} = \{4\}$
Addition rule	$\lvert S \cup T \rvert = \lvert S \rvert + \lvert T \rvert - \lvert S \cap T \rvert$	$\lvert \{3, 4\} \cup \{4, 5, 6\} \rvert = 2 + 3 - 1 = 4$

Counting Methods

Multiplication Principle:

Number of choices from sets $A_1, A_2, ..., A_n$:

$$|A_1| \times |A_2| \times ... \times |A_n|$$

Factorial:

$$n! = n \times (n-1) \times ... \times 1$$

$$0! = 1! = 1$$

$$n! = (n-1)! \times n$$

Permutations:

A data set of n objects has $n!$ permutations.

Combinations

Number of k-element subsets from n objects:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Example: 2-element subsets of {A, B, C, D, E}:

$$\binom{5}{2} = \frac{5!}{2!3!} = \frac{120}{2 \times 6} = 10$$

Probability

For equally likely outcomes:

$$P(\text{event}) = \frac{\text{number of favorable outcomes}}{\text{total number of possible outcomes}}$$

The probability $P(E)$ of an event E is a number between 0 and 1, inclusive. If all outcomes are equally likely:

$$P(E) = \frac{|E|}{|S|}$$

where $|E|$ is the number of outcomes in event E and $|S|$ is the total number of possible outcomes.

Concept	Formula
Not E	$P(\text{not } E) = 1 - P(E)$
E and F	$P(E \text{ and } F) = P(E \cap F) = P(E\|F) \times P(F)$
E or F	$P(E \text{ or } F) = P(E \cup F) = P(E) + P(F) - P(E \text{ and } F)$
Conditional Probability	$P(E\|F) = \frac{P(E \cap F)}{P(F)}$
Independent Events	E and F are independent if $P(E \text{ and } F) = P(E) \times P(F)$

If events E and F are independent, then $P(E|F) = P(E)$ and $P(F|E) = P(F)$.

Sequences

A **sequence** is an ordered list of elements, starting with a first element.

Example: The function $a(n) = n^2 + \left(\frac{n}{6}\right)$ with the domain of all positive integers $n = 1, 2, 3, ...$ defines an infinite sequence a_n where:

- $a_1 = 1^2 + \frac{1}{6} = \frac{7}{6}$
- $a_2 = 2^2 + \frac{2}{6} = \frac{13}{3}$
- $a_3 = 3^2 + \frac{3}{6} = \frac{19}{2}$
- and so on…

Arithmetic sequence: Each term differs from the previous by a constant d.

$$a_n = a_1 + (n-1)d$$

Geometric sequence: Each term is the previous term multiplied by a constant r.

$$a_n = a_1 \times r^{n-1}$$

8. Geometry Reference

Triangles

Pythagorean Theorem: For a right triangle with legs a, b and hypotenuse c:

$$a^2 + b^2 = c^2$$

Area: $A = \frac{1}{2} \times \text{base} \times \text{height}$

Sum of angles: Always $180°$

Circles

Circumference: $C = 2\pi r = \pi d$

Area: $A = \pi r^2$

where r is radius and d is diameter.

Rectangles

Area: $A = l \times w$

Perimeter: $P = 2l + 2w$

where l is length and w is width.

Rectangular Solids

Volume: $V = l \times w \times h$

Surface Area: $S = 2(lw + lh + wh)$

Cylinders

Volume: $V = \pi r^2 h$

Surface Area: $S = 2\pi r^2 + 2\pi rh$

where r is radius and h is height.

Remember: This reference sheet is for quick lookup during practice.

Understanding concepts deeply is more important than memorizing formulas.